COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 05

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Use logical equivalences to prove $\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \wedge q)$.

(b) **[2 points]** Let c mean *the battery is charged* and w mean *the device works*. Translate *charging the battery is sufficient for the device to work* and *working is necessary for being charged*. Do these statements rule out $w = T, c = F$?

(c) **[5 points]** Let m be a fixed odd integer. Prove that for every integer n, $n + m$ is even if and only if n is odd. Prove both directions from the definitions.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** Over $\mathbb{R}$, evaluate with justification: (i) $\forall x \exists y (y > x)$; (ii) $\exists y \forall x (y > x)$ (1 point each). Explain whether $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ are always equivalent (2 points).

(b) **[4 points]** All variables range over $\mathbb{R}$. Negate $\forall x (x > 0 \rightarrow \exists y (y > 0 \wedge \forall z (z > y \rightarrow P(x, y, z))))$. Leave no negation before a quantifier or compound formula.

(c) **[5 points]** Prove both equivalences by witness/arbitrary-element reasoning: (i) $\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$ (3 points); (ii) $\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$ (2 points).

Problem 3: Sets and functions [12 points]

(a) **[2 points]** Prove $A \cup (B \setminus A) = A \cup B$ by membership or set laws.

(b) **[3 points]** Describe an enumeration of all positive rationals that lists each rational exactly once. Explain why it reaches every rational and why the set is infinite.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x$, is injective.

(ii) The same function is surjective.

(iii) Changing that function's codomain to the even integers makes it bijective.

(iv) For arbitrary compatible functions, $f \circ g = g \circ f$.

(v) A function's range is always a subset of its codomain.

(vi) A relation from A to B must give exactly one output for each element of A.

(vii) Every bijection gives equal cardinalities for its domain and codomain.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** Seven distinct people stand in a line. A must be first and B must not be last. Count the lines.

(b) **[2 points]** Select an unordered four-person team from nine distinct people, then designate one member as captain. Count these teams with captains.

(c) **[2 points]** Count distinct strings formed by rearranging all letters of STATISTICS, with T as both the first and last letter. Equal letters are indistinguishable.

(d) **[2 points]** Eight distinct people sit around a circular table. B must not be immediately clockwise from A. Rotations are equivalent; reflections are different. Count the seatings.

(e) **[2 points]** Count length-six bit strings with four ones and first bit 1.

(f) **[2 points]** Choose eight cookies from five unlimited types, with exactly two of type A. Cookies of one type are identical, and order does not matter. Count the choices.

(g) **[2 points]** Find the coefficient of x^6 in $(x^2 + 2)^5$.

(h) **[2 points]** A string has length three or four over $\{A, B, C, D\}$, starts with A, and ends with a symbol other than A. Repetition is allowed. Count the strings.

Problem 5: Combinatorial proof [10 points]

(a) **[6 points]** For $n \geq 1$, give an explicit bijection between even-sized and odd-sized subsets of an n-element set, and prove that it is a bijection.

(b) **[4 points]** Deduce both identities below by counting subsets:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \qquad \sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0.$$

Problem 6: Inclusion-exclusion [9 points]

(a) **[6 points]** Use inclusion–exclusion to count permutations of $\{1, \ldots, 7\}$ with exactly two fixed points.

(b) **[3 points]** How many permutations of $\{1, \ldots, 7\}$ have exactly six fixed points? Explain.

Problem 7: Pigeonhole principle [7 points]

[**7 points**] Consider eleven team result records, each for three games against arbitrary opponents. The records need not form a league among these teams. A win earns three points, a draw one point, and a loss zero. Each record also gains a single bonus point if it contains at least one win. Prove that two records have the same total score.

Problem 8: Continuous and stopping models [6 points]

U has density $f(x) = 1/2$ on $[0, 2]$ and zero elsewhere. A separate variable N has density $g(x) = e^{-x^2/2}/\sqrt{2\pi}$. Independently repeated Bernoulli trials with success probability $1/4$ stop at the first success; T includes that trial.

(a) **[1 points]** Is $f(1) = 1/2$ the probability $P(U = 1)$? Give that probability.

(b) **[1 points]** Find $P(1/2 \leq U \leq 3/2)$ by interval area.

(c) **[1 points]** Name the distribution represented by g; no integration is required.

(d) **[1 points]** Find $P(T \geq 3)$.

(e) **[2 points]** State the countable stopping-outcome space and pmf of T, and verify normalization by a geometric series.

Problem 9: Knowledge and guessing [7 points]

One test item is selected. A student knows its answer with probability 1/2; with probability 1/3 she only eliminates two of its four options; otherwise she has no information. She answers known items correctly and otherwise guesses uniformly among the remaining options.

(a) [**3 points**] Find the probability of a correct answer.

(b) [**4 points**] Given a correct answer, find the posterior probabilities of all three knowledge states. Verify that they sum to one.

Problem 10: Collecting types [10 points]

Each draw independently produces one of $n \geq 2$ coupon types uniformly. Continue until all types are collected. Let G_i count the additional draws, including the new type, needed to move from i distinct types to $i + 1$.

(a) **[2 points]** Find $E(G_0)$ and explain.

(b) **[2 points]** For $0 \leq i < n$, identify the distribution of G_i and its mean.

(c) **[2 points]** Find the expected draws needed to obtain two distinct types, starting with none.

(d) **[4 points]** Derive the expected total draws needed for all n types. Explain why each stage retains a fixed success probability until it ends.

Draft Paper

Mock examination working space.

Draft Paper

Mock examination working space.